

Semi-Lagrangian DNS of turbulent channel flow: what the literature proves, what `slChannel` does, and where the gaps are

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Abstract

`slChannel` replaces the CFL-limited explicit advection of a staggered finite-difference channel DNS with unconditionally stable semi-Lagrangian (SL) advection, so that Δt is set by physical accuracy ($\Delta t^+ \lesssim 0.4$) rather than by $\text{CFL} \approx 0.28$. This report collects (i) what the semi-Lagrangian Navier–Stokes literature [?, ?, ?, ?] actually establishes, with the governing equations; (ii) what `slChannel` implements and has measured, including results that reproduce — and in two cases explain — phenomena the literature reported but did not analyse; (iii) the general problems of the approach; and (iv) the specific gaps that remain open and that the present campaign addresses. The headline: the time-integration skeleton (BDF2 along characteristics) has been settled since 1997–2001, and one turbulent channel was run with it in 2002 at $\Delta t^+ = 0.26$ [?]; what has never been published is the *statistics-fidelity map* of SL channel DNS — spectra, u_τ , and Reynolds stresses as functions of Δt^+ and Δx^+ against reference data — nor any stability theory for the step-decorrelated regime $\Delta t^+ \gtrsim 0.25$ where the wall-clock win actually lives.

1 Setting and notation

Incompressible Navier–Stokes in Lagrangian form,

$$\frac{d\mathbf{u}}{dt} = -\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0, \quad \frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}, t). \quad (1)$$

A semi-Lagrangian scheme advances the solution on a *fixed* grid by tracing, each step, the characteristic arriving at every grid node $\mathbf{x}_a$ backwards to its departure point (“foot”) $\mathbf{x}_d$, and interpolating the old field there. Wall units use u_τ and ν : $\Delta t^+ = \Delta t u_\tau^2 / \nu$, $\Delta x^+ = \Delta x u_\tau / \nu$. The `slChannel` operating points are the open channel at $\text{Re}_\tau \approx 182$ ($256^2 \times 100$, $\Delta x^+ = 7.7$) and the closed KMM [?] channel at $\text{Re}_b = 2792.8$ ($\text{Re}_\tau = 178.12$; $192^2 \times 256$ with $\Delta x^+ = 11.8$ and $256^2 \times 256$ with $\Delta x^+ = 8.8$). The Eulerian reference scheme (identical spatial discretization, AB2/CN IMEX) is capped at $\text{CFL} \approx 0.28$, i.e. $\Delta t^+ \approx 0.06$; the SL schemes run at $\Delta t^+ = 0.25$ – 0.40 .

2 What the literature proves

2.1 The premise: temporal over-resolution of Eulerian DNS

Xiu & Karniadakis [?] formalize why explicit advection wastes resolution in *time*. With $N \approx \text{Re}^{3/4}$ nodes per direction to resolve the Kolmogorov length, an eigenvalue scaling N^α of the discrete advection operator gives

$$\frac{\Delta t_{\text{CFL}}}{\tau_\eta} \propto \text{Re}^{1/2} \frac{1}{N^\alpha} \propto \text{Re}^{1/2-3\alpha/4}, \quad (2)$$

with $\alpha = 1$ (Fourier) to 2 (Chebyshev). At $\text{Re} = 10^4$ the CFL step sits one to four orders of magnitude *below* the Kolmogorov time: the smallest spatial scales are marginally resolved while the temporal scales are over-resolved by orders of magnitude. This is exactly the `slChannel` premise; at our $\text{Re}_\tau = 180$ operating point the physics limit $\Delta t^+ \approx 0.3\text{--}0.4$ (a fraction of the near-wall Kolmogorov time $\tau_\eta^+ \approx 2\text{--}3$, and of the sweeping-decorrelation time of the smallest resolved eddies) sits $5\text{--}7\times$ above the Eulerian cap.

2.2 The error structure of interpolated SL advection

The rigorous convergence analysis of Falcone & Ferretti [?] (anticipated by McDonald and the meteorological literature [?]) gives, for backward integration of order k and interpolation of polynomial order P ,

$$\frac{u^{n+1} - u_d^n}{\Delta t} = \frac{du}{dt} + \mathcal{O}\left(\Delta t^k + \frac{\Delta x^{P+1}}{\Delta t}\right). \quad (3)$$

Three consequences are proved and repeatedly verified in [?, ?, ?]:

1. **Non-monotonicity in Δt .** The error *decreases* as the step grows while the interpolation term dominates (fewer remaps per unit time), reaches a minimum, then grows as Δt^k . Xiu & Karniadakis map this competition explicitly (their Figs. 4–5: at intermediate spatial resolution the minimum sits at $\Delta t \approx 0.024$ for their cone problem; at high resolution pure Δt^2 growth remains).
2. **Small- Δt over-dissipation.** The $\Delta x^{P+1}/\Delta t$ term means an SL run at *small* step accumulates interpolation error at rate $1/\Delta t$ — Boukir et al.’s Remark 7(ii) [?]. SL should be run at large Δt or not at all.
3. **Low-order interpolation is unusable.** SL with linear interpolation is algebraically equivalent to first-order upwinding (excessively dissipative); cubic is the classical minimum [?]; Giraldo showed that for spectral-element interpolation of order $P \geq 4$ dispersion and dissipation errors essentially vanish at any Courant number [?].

Stability is subtle: L^∞ -stability holds for monotone (low-order) interpolation; for order > 2 on *equidistant* stencils von Neumann analysis shows instabilities can arise unless “sliding” stencils centred on the departure point are used (what everyone, including `slChannel`, does); for nonuniform grids “ L^2 stability holds although a rigorous theory is incomplete” [?]. Two further practical constraints from [?]: trajectories must not cross,

$$\Delta t < \|J_{\mathbf{u}}^{-1}\|, \quad J_{\mathbf{u}} = \partial \mathbf{u} / \partial \mathbf{x} \quad (4)$$

(in wall units at $\text{Re}_\tau = 180$ the near-wall shear gives $\Delta t^+ \lesssim \mathcal{O}(1)$, comfortably satisfied); and boundary feet “may search for points outside the domain due to the large time step” — the proposed remedy in the literature is subcycling grid points near the boundary with $\delta t < \Delta t$, acknowledged to complicate the implementation.

2.3 Second-order schemes: the Crank–Nicolson trap and the stiffly-stable cure

This is the core methodological content of [?] and [?], and it maps one-to-one onto the `slChannel` time schemes.

Backward integration. Both papers trace feet with the explicit second-order midpoint rule; writing $\alpha = \mathbf{x}_a - \mathbf{x}_d$,

$$\alpha = \Delta t \mathbf{u} \left(\mathbf{x}_a - \frac{\Delta t}{2} \mathbf{u}(\mathbf{x}_a, t^n), t^{n+\frac{1}{2}} \right), \quad (5)$$

iterated to convergence (the implicit midpoint variant gives “almost identical results”; the explicit form is preferred for velocity fields known only numerically [?]). `slChannel`’s iterated-midpoint departure solver is exactly this.

First order. Backward-Euler characteristics, $(u^{n+1} - u_d^n)/\Delta t = (-\nabla p + \nu \nabla^2 u)^{n+1}$: unconditionally stable, robust, first order — the direct analogue of `slChannel v1`.

The Crank–Nicolson trap. The obvious second-order scheme,

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}_d^n}{\Delta t} = -\nabla p^{n+1} + \nu \nabla^2 \left(\frac{\mathbf{u}^{n+1} + \mathbf{u}_d^n}{2} \right), \quad \frac{d\mathbf{x}}{dt} = \mathbf{u}^{n+\frac{1}{2}} = \frac{3}{2}\mathbf{u}^n - \frac{1}{2}\mathbf{u}^{n-1}, \quad (6)$$

“seems to be second-order in time ... however, our numerical experiments show that it is in fact still first-order” [?]: integrating (??) along the characteristic, the midpoint rule requires the *whole* right-hand side averaged between the departure point and the arrival point,

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}_d^n}{\Delta t} = \frac{1}{2} \left[(-\nabla p + \nu \nabla^2 \mathbf{u})_d^n + (-\nabla p + \nu \nabla^2 \mathbf{u})^{n+1} \right]. \quad (7)$$

Equation (??) is `slChannel`’s `v2` (explicit terms and the extrapolated pressure gradient averaged between departure and arrival, trajectory-CN diffusion). The paper then reports that the three-step splitting of (??) “develops a long-term instability ... due to the explicit part of the pressure term ... it could be treated by additional projections” — documented on the Taylor vortex at $\text{Re} = 10^6$ (their Figs. 9, 16), where the CN-type scheme drifts over hundreds of time units while the stiffly-stable scheme tracks the Eulerian reference. Independently, `slChannel` hit the same wall twice: the `pc` trajectory predictor diverged until the predictor was *projected* (one extra Poisson solve — literally the “additional projection”), and the `v2` incremental-pressure bookkeeping had to enter *before* the implicit solve to avoid degenerating to the non-incremental scheme.

The stiffly-stable (BDF2) scheme. Both [?] and [?] arrive at the same cure — replace the neutrally-stable trapezoidal combination by a backward-differentiation formula along characteristics:

$$\boxed{\frac{\frac{3}{2}\mathbf{u}^{n+1} - 2\mathbf{u}_d^n + \frac{1}{2}\mathbf{u}_d^{n-1}}{\Delta t} = (-\nabla p + \nu \nabla^2 \mathbf{u})^{n+1}}, \quad (8)$$

with *two independent feet*: $\mathbf{x}_d^n$ traced over one step Δt and $\mathbf{x}_d^{n-1}$ traced over two steps $2\Delta t$, both starting from the arrival node. Boukir et al. freeze one trajectory field $U^* = 2\mathbf{u}^n - \mathbf{u}^{n-1}$ for both feet and prove (Remark 4(i)) that *continuing the far foot from the near foot* (“bending”) drops the global order to one — the feet must be independent solutions of the same backward ODE. Xiu & Karniadakis trace the near foot with the extrapolated field $\mathbf{u}^{n+1/2} = \frac{3}{2}\mathbf{u}^n - \frac{1}{2}\mathbf{u}^{n-1}$ and the far foot with $\mathbf{u}^n$; either choice preserves second order (Taylor analysis). The three-step splitting used in [?, ?],

$$\frac{\hat{\mathbf{u}} - 2\mathbf{u}_d^n + \frac{1}{2}\mathbf{u}_d^{n-1}}{\Delta t} = 0, \quad (9)$$

$$\frac{\hat{\mathbf{u}} - \hat{\mathbf{u}}}{\Delta t} = -\nabla p^{n+1}, \quad \nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \hat{\mathbf{u}}, \quad \frac{\partial p}{\partial n} = -\nu \mathbf{n} \cdot [\hat{\mathbf{u}}/\Delta t + \nabla \times \boldsymbol{\omega}^{n+1}], \quad (10)$$

$$\frac{\frac{3}{2}\mathbf{u}^{n+1} - \hat{\mathbf{u}}}{\Delta t} = \nu \nabla^2 \mathbf{u}^{n+1}, \quad (11)$$

is non-incremental in the pressure with the rotational (Karniadakis–Israeli–Orszag [?]) boundary condition; the implicit solve and projection act with the effective step $\Delta t_{\text{eff}} = \frac{2}{3}\Delta t$. Boukir et al. prove stability under a non-restrictive step condition and error estimates of the type

$$\|\mathbf{u} - \mathbf{u}_h\| \leq C \left(\Delta t^2 + h^m + \frac{h^{m+1}}{\Delta t} \right) \quad (12)$$

for degree- m elements — second order in time, with the same interpolation-accumulation term as (?). `slChannel`’s `bdf2` scheme is the Boukir variant (frozen U^* , two independent feet, $\Delta t_{\text{eff}} = 2\Delta t/3$, non-incremental or incremental pressure), verified at self-convergence ratios 4.00 (incremental) and 2.0 (non-incremental, projection-splitting limited).

2.4 Strong vs. auxiliary form; interpolation choices

Xiu, Sherwin, Dong & Karniadakis [?] compare the *strong* form (backward particle tracking, as above) against the *auxiliary* form — solving the pure advection problem

$$\frac{\partial \tilde{\phi}}{\partial \tau} + \mathbf{u} \cdot \nabla \tilde{\phi} = 0, \quad \tilde{\phi}(\mathbf{x}, t^n) = \phi(\mathbf{x}, t^n), \quad \tau \in [t^n, t^{n+1}] \quad (13)$$

with an Eulerian substepper at its own CFL limit (the OIFS scheme of Maday, Patera & Rønquist [?]), so that $\phi_d^n = \tilde{\phi}(\mathbf{x}_a, t^{n+1})$ without any interpolation. Verdict: at intermediate steps the auxiliary form is preferable (no interpolation error, standard parallelization), but its cost grows with the number of substeps and it is *not* CFL-free; **at large time steps only the strong form is stable and its tracking cost is essentially independent of Δt** . `slChannel` is a strong-form code; this is the right branch for $\Delta t^+ \gtrsim 0.25$.

For the channel geometry specifically ([?], §3.3: Fourier $\times$ Fourier $\times$ spectral-element, i.e. structurally our grid), they reject global Fourier interpolation as “very high” cost and use *local* interpolation in the homogeneous directions — exactly the tensor-product-stencil choice of `slChannel` — comparing Lagrange stencils of orders 3–7 against *cubic Hermite* with the nodal derivatives computed spectrally (FFT differentiation): third-order Hermite matches seventh-order Lagrange accuracy at about half the cost, and retains exponential convergence where fixed-order Lagrange saturates. This is the closest published relative of `slChannel`’s prefiltered-spline `field_interp` path (C^2 B-spline in x, y via FFT symbol division + Hermite evaluation in z).

2.5 The one turbulence result: Xu, Xiu & Karniadakis (2002)

Reference [?] is the only published SL turbulent channel DNS in this line, and its parameters land startlingly close to ours:

	Xu et al. 2002 [?]	<code>slChannel</code> (2026)
Scheme	BDF2 characteristics, Eqs. (??)–(??)	same skeleton (<code>bdf2</code> , Boukir feet)
Discretization	Fourier $\times$ SE(Jacobi) $\times$ Fourier, 48^3	FD2 staggered, $192^2\text{--}768^2 \times 100\text{--}256$
Interpolation	<i>global</i> modal (all Fourier modes)	local tricubic/triquintic, Triton
Re_τ	116	178–182 (exact KMM Re_b)
Δt^+	0.2645 ($10\times$ Eulerian Δt)	0.25–0.40 ($4\text{--}7\times$)
Statistics window	$T = 20$ convective units	70 washouts (≈ 880 c.u., $10^4 t^+$)
Validation	Eulerian twin only, mean + rms	Eulerian twin + KMM/MKM data, profiles, stresses, spectra
Cost	$10\times/\text{step} \Rightarrow \text{net } 1.0\times$	$1.3\text{--}2.1\times/\text{step} \Rightarrow \text{net } 2\text{--}3.2\times$

Their findings: (i) the SL run is *stable at marginal resolution without de-aliasing* where the Eulerian twin is unstable at any Δt — interpolation supplies the missing high- k dissipation; (ii) mean velocity indistinguishable from the Eulerian reference “despite the very large time step”; (iii) *rms intensities differ, most visibly an elevated streamwise peak*, which they attribute to time-averaging error by comparing $T = 5$ vs $T = 20$ Eulerian averages; (iv) the global interpolation made the method $10\times$ slower per step, cancelling the $10\times$ step gain — they close by calling for “better local interpolation procedures” and report no follow-up channel study. Two details deserve emphasis. First, their $\Delta t^+ = 0.2645$ sits *just below* the two-foot stability boundary $\Delta t^+ \approx 0.30$ that `slChannel` mapped in 2026 — they were one courageous step-doubling away from a mysterious blow-up. Second, their elevated streamwise rms is the same signature `slChannel` measures with $44\times$ longer averaging (+5.6% peak u'_{rms}^+ at $\Delta x^+ = 11.8$); with converged statistics it is demonstrably *not* an averaging artifact.

3 What `slChannel` does

3.1 Scheme summary

Staggered MAC finite differences (x, y uniform periodic, z tanh-stretched), `torch.float64` infrastructure with fp32-accumulate Triton gather kernels for the interpolation. Feet by iterated midpoint (??) with trilinear (default) or tricubic trajectory sampling; fields remapped by tensor-product Lagrange stencils (tricubic $P=3$ / triquintic $P=5$; nonuniform-node weights in z against the actual grid, analytic inverse tanh map for stencil location).

The time scheme is Boukir BDF2 (??) and nothing else: frozen $U^* = 2\mathbf{u}^n - \mathbf{u}^{n-1}$, two independent feet at Δt and $2\Delta t$, update $\hat{u} = (4\bar{u} - \bar{u})/3$, $\theta=1$ z -solve and FFT projection at $\Delta t_{\text{eff}} = 2\Delta t/3$, with `noninc` (production) or `inc` pressure — velocity self-convergence ratios 2.0 and 4.00 respectively. Mass flux is imposed exactly by a uniform divergence-free shift (CaNS convention); Reynolds stresses are accumulated about the *time* mean at each component’s own staggered nodes.

The campaigns reported in §?? also exercised a first-order backward-Euler splitting (`v1`), a departure-averaged Crank–Nicolson scheme with incremental pressure (`v2`, self-convergence 3.93), projected predictor–corrector trajectories (`pc`, 3.78), and a C^2 prefiltered spline remap. All are reported here as *evidence*, and all were removed from the code once BDF2 won the stability-versus-accuracy trade: the released solver implements the single scheme above. An Eulerian IMEX path is retained purely as a like-for-like reference for validation and benchmarking.

3.2 One step in full: the discrete equations

This subsection writes out every substep as implemented in `solver.py/semilag.py`, in execution order. Notation: $V^n = (u, v, w)^n$ is the staggered MAC velocity (u at x -faces, v at y -faces, w at z -faces, one ghost layer per side; pressure at cell centres); G_h and D_h are the staggered gradient and divergence; L_{xy} and L_z the horizontal and wall-normal discrete Laplacians (second-order central, L_z on the stretched metric); $\mathcal{A}[\phi](\mathbf{x}_d)$ denotes the interpolation of a field ϕ at the departure points of the component grid on which ϕ lives. All ghost layers are rebuilt (*BC application*) after every operation that leaves them stale.

(1) **Trajectory velocity.** A single advecting field $\mathbf{U}^*$, defined per scheme:

$$\mathbf{U}^* = \begin{cases} \frac{3}{2}V^n - \frac{1}{2}V^{n-1} & \text{v1/v2, ab2 (target } t^{n+1/2}), \\ V^n & \text{traj_extrapolation: none (diagnostic, } \mathcal{O}(\Delta t)), \\ \frac{1}{2}(V^n + \tilde{V}^{n+1}) & \text{pc (projected predictor } \tilde{V}^{n+1}, \text{ see below),} \\ 2V^n - V^{n-1} & \text{bdf2 (frozen for both feet, target } t^{n+1}). \end{cases} \quad (14)$$

For `pc`, the predictor is a complete first-order step (SL advection of V^n by itself, plus the horizontal viscous tendency, plus the *lagged* pressure gradient $-G_h p^{n-1/2}$, then a backward-Euler z -solve) which is then *projected* — one extra Poisson solve. The projection is not optional: an unprojected predictor carries an $\mathcal{O}(\Delta t)$ divergence, SL trajectories through a divergent mid-velocity are compressive, and the scheme blows up ($u_\tau + 127\%$, §??).

(2) Departure points: iterated midpoint. For every arrival node $\mathbf{x}_a$ of each component grid, with $K = 2$ iterations (`n_traj_iters`):

$$\mathbf{x}_m^{(0)} = \mathbf{x}_a, \quad \mathbf{x}_m^{(s)} = \mathbf{x}_a - \frac{\Delta t}{2} \mathbf{U}^*(\mathbf{x}_m^{(s-1)}), \quad s = 1, \dots, K, \quad \mathbf{x}_d = 2\mathbf{x}_m^{(K)} - \mathbf{x}_a, \quad (15)$$

i.e. $\mathbf{x}_d = \mathbf{x}_a - \Delta t \mathbf{U}^*(\mathbf{x}_m^{(K)})$: the explicit iterated midpoint rule (??). $\mathbf{U}^*$ is sampled at the (staggered) midpoints by trilinear interpolation on the fast path (`traj_order: 2`; its C^0 interpolant caps *self*-convergence at fixed grid at $\mathcal{O}(\Delta t)$ with a small h^2 coefficient — tricubic sampling, `traj_order: 4`, restores clean $\mathcal{O}(\Delta t^2)$, ratio 3.93). x, y are unwrapped modulo the period; z is clamped to the wall-adjacent node range and every clamp is counted (`clamped` diagnostic; measured 0 over $10^4 t^+$ in the closed channel). For `bdf2` the same $\mathbf{U}^*$ traces *two independent* feet from the same arrival node, (??) with Δt and with $2\Delta t$ — never the far foot continued from the near one (Boukir Remark 4(i)).

(3) Remap. The old field is evaluated at the feet by a tensor-product Lagrange stencil of $P+1$ nodes per direction ($P = 3$ tricubic, $P = 5$ triquintic):

$$\mathcal{A}[\phi](\mathbf{x}_d) = \sum_{i,j,k} w_i^x(x_d) w_j^y(y_d) w_k^z(z_d) \phi_{ijk}^n, \quad w_i(\xi) = \prod_{m \neq i} \frac{\xi - \xi_m}{\xi_i - \xi_m}. \quad (16)$$

In x, y the nodes are uniform and the weights close-form; in z the weights are built *against the actual stretched nodes* z_c/z_f (uniform- ξ weights on the tanh image would silently lose an order: z_c are face midpoints, not the tanh-map image), the stencil is located by the analytic inverse tanh map plus one node compare (no search), sliding stencils centred on the foot in the interior, one-sided at the walls. The index/weight set of (??) is computed once per component and step and reused for every field remapped on that grid — this is what makes the `v2` departure-evaluated right-hand sides in (??) nearly free. The alternative C^2 path (`field_interp: spline`) replaces (??) by a prefiltered cubic B-spline in x, y (FFT symbol division) and a nonuniform C^2 cubic spline in z with Hermite evaluation.

(4) Explicit horizontal tendency. On V^n , $R_{xy} = \nu L_{xy} V^n$ (no body force here — the flux constraint is substep 8). `v2` time-centres it by AB2, $R_{xy}^{n+1/2} = \frac{3}{2} R_{xy}^n - \frac{1}{2} R_{xy}^{n-1}$; `bdf2` extrapolates to t^{n+1} , $R_h = 2R_{xy}^n - R_{xy}^{n-1}$ (`bdf2_xy_rhs: extrap`; `lagged` uses R_{xy}^n).

(5) Advected predictor. Per scheme:

$$\text{v1: } u^* = \mathcal{A}[V^n](\mathbf{x}_d) + \Delta t R_{xy}, \quad (17)$$

$$\text{v2: } u^* = \mathcal{A}[V^n](\mathbf{x}_d) + \Delta t \left[\frac{1}{2} (\mathcal{R}_d + \mathcal{R}_a) + \frac{1}{2} (\nu L_z V^n)_d \right], \quad \mathcal{R} = R_{xy}^{n+1/2} - G_h p_{\text{ext}}, \quad (18)$$

$$\text{bdf2: } u^* = \frac{4}{3} \mathcal{A}[V^n](\mathbf{x}_d^{\Delta t}) - \frac{1}{3} \mathcal{A}[V^{n-1}](\mathbf{x}_d^{2\Delta t}) + \Delta t_{\text{eff}} [R_h - G_h p_{\text{ext}}], \quad \Delta t_{\text{eff}} = \frac{2}{3} \Delta t. \quad (19)$$

In (??) subscripts d/a mean the field evaluated at the departure point (via the reused stencil of substep 3, with BC-consistent ghosts so near-wall feet interpolate meaningful values) and at the arrival node: the *entire* explicit right-hand side, including the extrapolated pressure

gradient $p_{\text{ext}} = 2p^{n-1/2} - p^{n-3/2}$, is averaged along the characteristic — this is exactly what distinguishes the true second-order scheme (??) from the first-order trap (??). The z -viscous term enters with its departure half only; its arrival half is the implicit solve below. In (??), $p_{\text{ext}} = 2p^n - p^{n-1}$ (full levels) for `bdf2_pressure: inc` and is omitted for `noninc`; dividing Boukir’s update (??) by the BDF2 leading coefficient $\frac{3}{2}$ is what produces the $\frac{4}{3}/\frac{1}{3}$ foot weights and the effective step Δt_{eff} carried by every subsequent substep. The pressure term must enter *here*, before the implicit solve: adding it after is algebraically identical to the non-incremental scheme (verified analytically and numerically).

(6) Implicit wall-normal diffusion. One tridiagonal solve per horizontal position and component (θ -scheme on the stretched metric, no-slip rows at the walls):

$$(I - \theta \delta t \nu L_z) u^{**} = u^* + (1 - \theta) \delta t \nu L_z u^*, \quad (\theta, \delta t) = \begin{cases} (\frac{1}{2}, \Delta t) & \text{v1 (CN at arrival),} \\ (1, \frac{\Delta t}{2}) & \text{v2 (arrival half of trajectory-CN),} \\ (1, \Delta t_{\text{eff}}) & \text{bdf2.} \end{cases} \quad (20)$$

(7) Projection. With $\delta t = \Delta t$ (v1/v2) or Δt_{eff} (bdf2),

$$L_h \phi = \frac{D_h \mathbf{u}^{**}}{\delta t}, \quad \mathbf{u}^{n+1} = \mathbf{u}^{**} - \delta t G_h \phi, \quad p^{\text{new}} = \begin{cases} \phi & \text{v1, bdf2-noninc,} \\ p_{\text{ext}} + \phi & \text{v2, bdf2-inc.} \end{cases} \quad (21)$$

The Poisson solve diagonalizes x, y by real FFTs with the second-order modified wavenumbers $\tilde{k}^2 = 2(1 - \cos k\Delta)/\Delta^2$, leaving one tridiagonal system in z per mode pair (k_x, k_y) with homogeneous Neumann ends (built into the staggered stencil — no explicit pressure BC, hence no rotational-BC machinery of (??) is needed); the singular Neumann–Neumann $(0, 0)$ mode is pinned. On the incremental branches the splitting error of (??) is $\mathcal{O}(\Delta t^2)$ (self-convergence ratios 3.93/4.00); the non-incremental branch is first-order in the projection but robust (ratio 2.0).

(8) Exact bulk-flux constraint. Last, the CaNS-convention mass flux enforcement:

$$c = U_b - \langle u^{n+1} \rangle_V, \quad u^{n+1} += c \text{ (uniformly)}, \quad f = c/\Delta t, \quad (22)$$

where $\langle \cdot \rangle_V$ is the volume average. A uniform shift is divergence-free, so it commutes with the projection just performed; the driving force f is *diagnosed* from the correction rather than steered towards it. This replaces an integral controller whose error dynamics were an undamped oscillator with $1/\Delta t$ gain — under which any Δt sweep partly measured the controller instead of the scheme.

Bootstrapping. `bdf2` assumes constant Δt (the $2\Delta t$ foot and the $\frac{4}{3}/\frac{1}{3}$ weights); on the first step, after a restart, or on any Δt change the step degenerates to one BDF1 step (backward-Euler characteristics: $\mathbf{U}^* = V^n$, single foot, $\theta = 1$ at full Δt) which seeds the V^{n-1} and R_{xy}^{n-1} histories — one $\mathcal{O}(\Delta t^2)$ local error, the same policy as the documented AB2 restart re-bootstrap. `bdf2` configs pin Δt (`dt_update_interval: 0`).

3.3 Reproductions of literature phenomena

1. **The CN long-term fragility** ([?], their Figs. 9/16): observed as the pc-without-projection blow-up ($u_\tau + 127\%$) and cured exactly as the paper suggests, by one extra projection per step.

2. **Non-monotonic error / the “sweet spot”** (Eq. ??): the M3 campaign finds statistics *degrade* both above $\Delta t^+ \approx 0.22$ (floor, below) and, in over-dissipation terms, at small Δt^+ — SL at the Eulerian Δt is strictly worse than Eulerian, consistent with Remark 7(ii) of [?].
3. **BDF2 damping vs. CN neutrality**: `bdf2` removes the `v2` trajectory-CN’s neutral high- k remap behaviour, as the stiffly-stable construction intends [?, ?].
4. **Second-order convergence** of all three second-order variants verified by Richardson self-convergence with frozen forcing, matching the Taylor-analysis claims of both papers.

3.4 Results beyond the literature

(a) **The high- k spectral floor and its mechanism.** For $\Delta t^+ > \Delta t_*^+ \approx 0.22$ (streamwise CFL ≈ 0.65 , the sweeping-decorrelation time of the highest resolved modes), the streamwise spectrum at $z^+ \approx 15$ develops a tail floor 20–100 $\times$ above the Eulerian reference. The mechanism is *not* interpolation error: quintic interpolation, C^2 splines, and fourth-order trajectory sampling all *raise* the floor (interpolation error is the floor’s *dissipation*); the injection is the AB2 *trajectory-velocity extrapolation* $\frac{3}{2}\mathbf{u}^n - \frac{1}{2}\mathbf{u}^{n-1}$, which amplifies content that decorrelates within one step. Setting the extrapolation off kills the floor at the cost of first-order trajectories. Nothing in [?, ?, ?, ?] anticipates this: their analyses assume smooth fields with bounded Lagrangian derivatives, an assumption turbulence at $\Delta t^+ \gtrsim 0.2$ violates by construction.

(b) **The stochastic two-foot instability.** Model the highest resolved modes as phase-decorrelated between steps (random departure-phase noise). A multistep SL update $\sum_i a_i \bar{u}_i$ built from independently traced feet then has per-step tail-energy gain $\sum_i a_i^2$, because decorrelated phases add in energy, not amplitude. For the BDF2 combination $\hat{u} = \frac{4}{3}\bar{u} - \frac{1}{3}\bar{\bar{u}}$,

$$G_{\text{BDF2}} = \left(\frac{4}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \frac{17}{9} \approx 1.89, \quad G_{\text{single foot}} = 1. \quad (23)$$

When G exceeds the per-step viscous-plus-interpolation damping of the tail, the floor stops being a floor and becomes a monotonic runaway. A 1-D SL model (cubic remap, AR(1) velocity decorrelation) predicts the bifurcation; the GPU campaign confirms it: `bdf2` is clean at $\Delta t^+ = 0.25$ and blows up monotonically at $\Delta t^+ \geq 0.30$, while single-foot schemes (gain 1) have bounded floors at all tested $\Delta t^+ \leq 0.4$. This is, to our knowledge, a novel result — it implies second-order multistep characteristics schemes carry an *intrinsic* large- Δt^+ stability tax that no deterministic truncation analysis reveals, and it explains why [?] at $\Delta t^+ = 0.2645$ saw nothing.

(c) **Converged statistics against reference data.** 70-washout closed-channel runs at the exact KMM $\text{Re}_b = 2792.8$: at $\Delta x^+ = 11.8$ the SL `bdf2` scheme delivers $\text{Re}_\tau = 173.9$ vs. MKM’s 178.12 (−2.4%), where the Eulerian twin on the *same grid* gives +0.9% — i.e. the SL interpolation itself costs $\approx 3\%$ in u_τ at this resolution (only $\approx 1\%$ at $\Delta x^+ = 7.7$ in the open channel), alongside a +5.6% elevated streamwise-rms peak and a low w' buffer layer. This is the quantitative version of what [?] glimpsed and set aside. The resolution-dependence hypothesis has since been confirmed: the same scheme at $256^2 \times 256$ ($\Delta x^+ = 8.8$, identical z -grid, 70 washouts) recovers $\text{Re}_\tau = 176.4$ (−1.0%) and reduces the streamwise-rms excess to +3.4% — the deficit is a wall-parallel-resolution effect of the interpolation dissipation, not a systematic bias of the time scheme, and the z -grid ($\Delta z^+ = 0.36$ at the wall, 2.4 at the centreline) is not implicated.

(d) **Wall-clock accounting on modern hardware.** With register-resident Triton gather kernels the full three-component tricubic advect costs 126 ms at $768^2 \times 180$ (fp32-accumulate-64); the SL step is 1.3–2.1 $\times$ the Eulerian step cost, and at the physics-limited Δt^+ the net gain is

2–3.2 $\times$ per simulated time unit. Measured end-to-end on the production closed-channel case at $256^2 \times 256$ (triquintic, $\Delta t^+ = 0.25$): 336 ms/step against the Eulerian twin’s 203 ms/step at $\Delta t^+ \approx 0.057$, i.e. 15.6 versus 41.4 s per simulated time unit, a net **2.65 $\times$** — where [?] reported net 1.0 $\times$ (global interpolation) and [?] 4 $\times$ on laminar spectral-element cases. The remaining cost is shared fp64 infrastructure (FFT-Poisson, implicit z -diffusion), not the SL machinery.

4 The problems, in general

1. **Interpolation dissipation is the effective subgrid model.** Equation (??)’s $\Delta x^{P+1}/\Delta t$ term is not a small perturbation in DNS: it is a scale-selective dissipation concentrated exactly where the near-wall cycle lives, and it biases u_τ and the rms profiles. The literature proves the error term; it never measured its effect on turbulence statistics. Every “better interpolant” trades floor height against dissipation — it cannot fix both, because the injection (item 2) is a different mechanism.
2. **Extrapolated trajectories amplify step-decorrelated content.** Both the bounded AB2 floor and the two-foot runaway (??) are stochastic phenomena of the $\Delta t^+ \gtrsim 0.2$ regime. All published SL-NS stability theory is deterministic and smooth; it is silent here.
3. **Pressure/projection splitting trades order against robustness.** The exactly-second-order routes (incremental pressure, departure-averaged CN) are precisely the fragile ones ([?]'s long-term instability; our pc history); the robust non-incremental BDF2 caps the velocity self-convergence near first order in the projection. This tension is intrinsic, not an implementation artifact.
4. **Boundary feet and symmetry planes.** Feet leaving the domain are rare in a channel ($w \rightarrow 0$ at walls; measured `clamped` = 0 over $10^4 t^+$), but the *interpolation stencil* near a boundary is one-sided and symmetry-blind: at a free-surface/symmetry plane the correct extension is a mirror (u, v even, w odd), and clamping/one-sided stencils get w wrong at first order. This corrupted the open-channel near-surface v, w statistics and motivated the closed-channel pivot; the literature’s only remedy (near-boundary subcycling [?, ?]) has never been tested in turbulence.
5. **No discrete conservation.** Advective-form SL conserves neither momentum nor energy; `slChannel` pins the bulk flux exactly (first moment) but second moments drift at the interpolation-dissipation level. None of [?, ?, ?, ?] addresses conservation at all.
6. **Cost is decided by the interpolation implementation.** The 2002 result died on a 10 $\times$ interpolation bill. The GPU changes the economics — gathers and fp32 tensor arithmetic are cheap, bandwidth-bound fp64 stencils are not — but only with kernels designed for it.

5 Where the gaps are

1. **The statistics-fidelity map (being filled by this campaign).** No published work reports spectra, u_τ , and Reynolds-stress budgets of an SL channel DNS against reference data as functions of Δt^+ and Δx^+ . The M3 sweep (Δt^+ axis), the KMM comparison ($\Delta x^+ = 11.8$), and the running tc256 case ($\Delta x^+ = 8.8$) constitute exactly this map. Publishable as-is once tc256 lands.
2. **Stochastic stability theory for characteristics schemes.** The ℓ^2 -gain argument (??) generalizes: any multistep foot combination has gain $\sum a_i^2 \geq 1$ with equality only for

single-foot schemes; a θ -blended single-foot scheme with BDF2-like damping but $G \leq 1$ would push the boundary past $\Delta t^+ = 0.3$. Design space untouched in the literature.

3. **Interpolation as a quantified LES model.** $u_\tau(\Delta x^+, P, \text{scheme})$ — the -3% vs -1% resolution branch is in progress; the order branch (quintic at $\Delta x^+ = 11.8$) and the Hermite-with-spectral-derivatives variant of [?] (never tried on a staggered FD grid; our C^2 spline is its cousin) are cheap follow-ups.
4. **Symmetry-consistent boundary treatment.** Mirror extension of the stencils at a free-slip plane (u, v even, w odd) vs. the current clamping; would reopen the cheaper half-channel configuration with clean near-surface statistics.
5. **Conservative SL for wall turbulence.** Flux-form/remap SL (SLICE-class) has never met a wall-bounded DNS; it would remove the second-moment drift at its root.
6. **Higher Re_τ and the economics at scale.** The premise (??) strengthens with Re ; the Re_{550} A/B benchmark ($768^2 \times 180$) is staged in the repository, and the infrastructure (PCR Poisson, fused z -Laplacian) is where the remaining wall-clock lives.

6 Parallelisation

`slChannel` is a single-GPU code today; this section separates what already parallelises (and how) from what a multi-GPU version would have to solve. The short version: the SL machinery is embarrassingly parallel within a device, and across devices it needs *wider but still static, neighbour-only* halo exchanges — the genuinely non-trivial communication of the step is the FFT Poisson solve, which SL inherits unchanged from any pencil-decomposed channel code.

6.1 Within one device

Every substep of §?? is data-parallel over grid points with no global iteration: (1)–(2) are pointwise linear combinations and short gather chains; (3) is a tensor-product gather — for each arrival node, $3(P+1)$ one-dimensional weights and a $(P+1)^3$ -point read; (6) is a batch of $n_x n_y$ *independent* tridiagonal systems per component ($65,536$ at 256^2 — far more parallelism than the GPU needs); (7) is a pair of 2-D FFTs plus another independent tridiagonal batch; (8) is a reduction and a broadcast. Two facts make the gather fast in practice. First, the departure map is a smooth deformation of the grid: neighbouring arrival nodes have neighbouring feet, so the loads are near-coalesced and cache behaviour is Eulerian-like, not scatter-like. Second, the weight tensors never need to exist in memory: the hand-written Triton kernels keep the $3(P+1)$ weights per point in registers and fuse trajectory iteration, stencil location and gather into one pass ($126/143$ ms tricubic/quintic for all three components at $768^2 \times 180$), where a graph compiler that materializes $(N, P+1)$ weight tensors pays ~ 10 GB of traffic for the same arithmetic. Precision follows the hardware: flop-dense interpolation in fp32 with fp64 accumulation, bandwidth-bound stencils and the whole pressure path in fp64.

6.2 Decomposition choice

The channel fixes the decomposition almost by itself. The z direction is non-periodic, stretched, carries the implicit solves (full-column tridiagonals) and the wall boundary conditions — so z stays on-device: decompose in x and/or y (“ z -pencils”). Then substeps (1), (2), (4), (5), (6), (8) are local up to standard halo exchanges in x, y ; the walls are never split. The Poisson solve needs full x -lines, then full y -lines, then full z -columns: the standard transpose cycle (z -pencils $\rightarrow x$ -pencils $\rightarrow y$ -pencils $\rightarrow z$ -pencils) of every 2DECOMP-class channel code (CaNS included).

SL adds nothing to it — the projection operates on fields at arrival nodes and never sees a departure point.

6.3 The SL-specific part: wide, anisotropic, static halos

What SL changes is the *width* of the neighbour exchange. An Eulerian second-order stencil needs a one-cell halo. An SL foot lies up to $\text{CFL}_x^{\max}$ cells upstream, and the interpolation stencil extends $\lceil P/2 \rceil$ nodes beyond it; the trajectory iterations sample $\mathbf{U}^*$ at midpoints inside the same envelope. Per direction,

$$h = m \lceil \text{CFL}^{\max} \rceil + \lceil P/2 \rceil, \quad m = \begin{cases} 1 & \text{single foot (v1, v2, pc),} \\ 2 & \text{far foot of bdf2 (depth } 2\Delta t\text{).} \end{cases} \quad (24)$$

At the production operating point (trajectory $\text{CFL} \approx 3$, measured $\max \text{CFL} \approx 2.7$ at $\Delta t^+ = 0.25$): tricubic **v2** needs $h = 5$; quintic **bdf2** needs $h = 2 \cdot 3 + 3 = 9$ in x . Three properties keep this benign. First, the halo is *static and known at step start*: one cheap global $\max |\mathbf{u}|$ bounds every foot, so the exchange is a regular neighbour transfer of fixed width — no dynamic, data-dependent communication, no owner-lookup of feet, in contrast to SL on unstructured meshes where foot location is itself a parallel search problem (the NWP literature’s on-demand thin-halo machinery [?] is overkill here). Second, it is *anisotropic in the right way*: the mean flow sweeps in $+x$, so feet lie predominantly upstream in x (CFL_y and CFL_z are several times smaller), and the wide halo is needed on one axis, mostly one side. Third, the exchange serves *all* gathers of the step: **bdf2** needs halos of three fields (V^n , V^{n-1} , $\mathbf{U}^*$ — and $\mathbf{U}^*$ is a local linear combination of the first two, so effectively two), and **v2**’s departure-evaluated right-hand sides ride the same stencils.

The volume arithmetic works out. At $h = 9$ on a 256^2 -per-GPU z -pencil the halo is $\sim 7\%$ of the subdomain per exchanged field — and the SL step runs at $4\text{--}7\times$ the Eulerian Δt , so per simulated time unit the SL scheme moves a halo volume *comparable* to the Eulerian twin’s thin-but-frequent exchanges. The hard scaling limit is geometric: strong scaling stops being useful when the local extent approaches $\sim 2h$ (≈ 20 cells at quintic **bdf2**), which for the KMM box means $\mathcal{O}(100)$ GPUs before the SL gathers even notice — the transpose all-to-all of the Poisson solve will saturate long before that, exactly as in Eulerian pencil codes.

6.4 Overlap and ordering

The step order of §?? is overlap-friendly: $\mathbf{U}^*$ is formed first (substep 1) and its halo posted asynchronously; departure points and remap of the *interior* (feet whose stencils stay on-device — the bulk, since $h \ll n_{\text{local}}$) proceed while boundary-zone halos are in flight; the boundary-zone gathers run last. The tridiagonal batches need no communication at all in z -pencils. The reductions of substep (8) and the $\max |\mathbf{u}|$ bound are single all-reduce scalars. None of this is implemented — but none of it is novel either: it is the standard structured halo-overlap pattern, with (??) as the only SL-specific input. The honest summary: parallelising SL channel DNS is *not* the hard part of this method; the two-foot stability boundary (??) and the interpolation-dissipation bias are.

Note on tooling

Parts of the implementation, the analysis scripts and the drafting of this report were carried out with the assistance of Claude (Anthropic). All results reported here were produced by the code in the repository and verified by the author, who takes full responsibility for their content.

7 Verdict

Boukir et al. supply the theorem-backed time skeleton; Xiu & Karniadakis supply the splitting that survives long integrations, the error structure, and the warning about Crank–Nicolson; Xu et al. prove an SL turbulent channel runs stably at $\Delta t^+ \approx 0.26$ and leave its statistics unexamined; Xiu et al. 2005 settle strong-vs-auxiliary in favour of the strong form at large steps and sketch the local-interpolation path our code takes. Everything downstream of that — the spectral floor and its extrapolation mechanism, the 17/9 two-foot bifurcation, the interpolation-dissipation cost in u_τ measured against MKM with converged statistics, and the GPU cost model that finally makes the method a net win — is not in the literature. Those are the contributions, and the fidelity map is now closed on both axes. The resolution branch: tricubic at $\Delta x^+ = 11.8$ gives $\text{Re}_\tau = 173.9$ (−2.4%), at $\Delta x^+ = 8.8$ $\text{Re}_\tau = 176.4$ (−1.0%). The interpolation-order branch: triquintic at $\Delta x^+ = 8.8$ recovers Re_τ to +0.4% of MKM with the peak u'_{rms}^+ within −0.2% — inside sampling noise, and better than the Eulerian twin on the same grid — for +2.6% on the step cost. Interpolation dissipation is therefore the entire statistics bias of the method, and it is removable by order and resolution rather than intrinsic. What remains intrinsic is temporal: the extrapolation-driven spectral floor above $\Delta t^+ \approx 0.22$ and the 17/9 two-foot gain that ends the usable range at $\Delta t^+ \approx 0.30$.

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